

Generative AI in music learning: A mixed-method study of skill development, AI dependence, and the moderating role of AI literacy

Yanran Ren^{a,*}, Safeer Ullah Khan^b

^a School of Tourism Management, Wuhan Business University, 430056, Wuhan, China

^b College of Management, Shenzhen University, Shenzhen, China

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ABSTRACT

This study examines how generative AI (Gen-AI) is associated with music learning outcomes by shaping underlying cognitive and self-regulatory mechanisms. Drawing on Self-Regulated Learning (SRL) and Cognitive Load Theory (CLT), the study investigates how students' use of Gen-AI is associated with music-specific domain skills and dependence on AI tools, and how these dual pathways are associated with the quality of musical performance. It further explores the moderating role of AI literacy in shaping these associations. A mixed-method approach was employed. Quantitative data were collected from 1142 music students across six leading university music schools in China using an online questionnaire and analyzed through Covariance-Based Structural Equation Modeling (CB-SEM). In addition, qualitative data were obtained from 15 experts in music education and AI-assisted music practice and analyzed using thematic analysis. The quantitative results show that Gen-AI use is positively associated with domain skills while also being positively associated with dependence on AI tools. Both domain skills and dependence are significantly associated with performance quality. Mediation analysis indicates that Gen-AI is indirectly associated with performance through a dual pathway: a positive association via skill development and a negative association via AI dependence. Moreover, AI literacy significantly moderates the associations between Gen-AI use, domain skills, and dependence, strengthening the positive pathway while weakening the negative one. The qualitative findings support and extend these results by highlighting how Gen-AI facilitates skill development, contributes to cognitive offloading, and how AI literacy functions as a meta-cognitive regulatory capability in guiding effective AI use. The study contributes to theory by extending the application of SRL and CLT to AI-mediated creative learning environments and by highlighting the dual cognitive mechanisms through which Gen-AI is associated with learning outcomes. Practically, the findings provide guidance for integrating AI in music education while emphasizing the importance of developing AI literacy to support balanced and effective AI use. Because the design is cross-sectional, the reported relationships are interpreted as associations rather than causal effects.

1. Introduction

The use of artificial intelligence (AI) in the education industry has been associated with substantial changes, especially with generative AI (Gen-AI) in artistic fields like music. Gen-AI systems have the ability to produce melodies, produce accompaniments, improvise in any style, and can provide instantaneous feedback. These systems can now be accessed freely by students through such websites as AIVA, Soundraw, Boomy and AI assisted digital audio stations (Pram & Morreale, 2025). These systems can support interactive and individualized learning, in which students have fewer barriers in learning music and more rapidly attain

musical competencies (Sanganeria & Gala, 2024). However, certain risks associated include overreliance, lowering self-esteem, and dilution of significant music skills when students let AI do cognitive and creative tasks (Shuang Zhang et al., 2024). In this study, such risks are conceptualized as a form of dependence on AI tools, defined not merely as frequent use, but as a tendency to substitute internal cognitive and creative processes with externally generated outputs. This distinction is important, as strategic use of AI may support learning, whereas dependence reflects a reduced engagement in self-regulated practice. As Gen-AI enters the field of music education, it is essential to understand not only how it is associated with learning but also how and when it helps or

* Corresponding author.

E-mail addresses: renyanran@wbu.edu.cn (Y. Ren), safeer89@outlook.com (S.U. Khan).

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hinders a student's musical learning.

Previous studies on technology in music education have focused on instrumental teaching systems, digital notation, or web-based pedagogy, with limited attention to the unique affordances and risks of generative models. Unlike static tools, Gen-AI systems are engaged in co-creative activities, and they tend to function as intelligent collaborators. The shift from passive tools to active agents alters psychological and pedagogical dynamics. In some cases, Gen-AI can support learning by way of direct feedback, model style norms, and exploratory practice building (Wang & Liu, 2025). Harmonic progressions created by AI can aid in internalizing voice-leads, and real-time pitch-correcting programs can help with intonation when teaching for voice training (Sanganeria & Gala, 2024). Such affordances indicate a virtuous pathway in which use of Gen-AI is associated with improvements in music-specific domain skills, such as aural perception, theoretical understanding, technical skill, and improvisational skill (Ma & Wang, 2025). Either the simplicity or the complexity of AI-generated content may incite cognitive offloading, whereby students use external systems to perform tasks that they otherwise would practice internally (Rahman et al., 2025). This process, termed technology dependence, has manifested in learning environments with calculators, grammar checkers, and navigation systems (Dron, 2022). In music, however, overuse of AI-leveraged tools, such as engaging AI to generate melodies or correct rhythms, may work against the effective development of procedural memory, creative agency, and resilience in performance. Students may gain momentum in making use of AI responses, but may lack in the ability to solve independent musical problems, especially in the areas of unstructured and high-pressure performance (Frater & Mushininga, 2025). Hence, while Gen-AI may enhance some abilities, it may be associated with increased dependency that compromises long-term musical independence and performance standards.

Such an ambivalence is reflective of the necessity of a dual pathway approach that takes into account the enabling and the disabling potential that Gen-AI technology may hold for music education. It is essential within such an approach to differentiate between skills acquired and dependence on a technological tool as mediating mechanisms between the application of the Gen-AI and its performance outcomes. One important consideration in this regard is the concept of AI literacy, defined as the ability to understand, critically evaluate, and ethically use AI technologies. Students who have high levels of AI literacy tend to engage with the application of the Gen-AI technology in a reflective manner, acknowledging its shortcomings and harnessing it in a strategic manner as part of the practice rituals (Ayyoub et al., 2025). The application of AI could thus be used by such students in a targeted manner, without falling into the trap of dependency. Low AI literacy, on the other hand, could cause users to accept AI output as authoritative and thus reduce self-regulation (Shahzad et al., 2025). Thus, AI literacy may shape both the benefits and risks associated with Gen-AI use.

AI literacy cannot be viewed in terms of digital literacy in its broadest sense. Digital literacy refers to the practical skills of accessing and using digital devices and software (e.g., managing files, searching for information, or ICT competences). AI literacy is a higher-order, domain-specific capability that involves understanding how generative systems produce probabilistic outputs, assessing the credibility and originality of those outputs, and reasoning about the ethical and pedagogical consequences of delegating cognitive or creative work to a machine. In this study, AI literacy is therefore conceptualized as a metacognitive and self-regulatory skill which influences how students regulate their cognitive resources while engaging with the Gen-AI system, rather than a constant predictor of learning performance, intermediary variable, or equivalent to technological ability per se.

Although there is growing research interest in the use of AI in education, little empirical research has been conducted into the effects of artificial intelligence on music learning, and no previous research has tested a comprehensive model encompassing skill development, dependence and AI literacy as predictors of performance quality. Most of

the available literature is largely descriptive or developmental in nature rather than outcomes for users (Southworth et al., 2023; Shan Zhang et al., 2025). Moreover, the conceptual framework that defines AI literacy in the music domain is also underresearched, often conflated with general digital literacy or technical proficiency (Sanganeria & Gala, 2024). A theoretically grounded investigation is required to provide insights into the relationship between Gen-AI and music and when its use will prove to be most effective.

This study is based on two theories: Self-Regulated learning (SRL) theory (Zimmerman, 2002) and Cognitive load theory (CLT) (Sweller, 1988) to conceptualize the relationship between Gen-AI and music learning processes. According to SRL, learning is optimal when students are able to set goals, monitor progress, and regulate cognitive and motivational strategies. With strategic deployment, gen-AI tools can function as external regulators, triggering self-regulation, offering real-time feedback, and scaffolding practice, improving metacognitive control and motivation (Nguyen & Barbieri, 2025). However, without sufficient self-regulatory competence, learners may become overly reliant on AI, displacing internal cognitive processes and undermining the development of autonomous musical agency. Cognitive Load Theory helps to make sense of this process by distinguishing between germane, intrinsic, and extraneous cognitive load: well-designed AI support can reduce extraneous load (e.g., by automating notation or suggesting harmonies), allowing for the learning of the skills and creative improvisation in working memory (Sweller, 2010). In contrast, poorly integrated or cognitively overwhelming AI use may contribute extraneous load or cause cognitive offloading that can harm schema development and long-term retention. Based on these theories and gaps, this research responds to the following questions:

RQ1: Is Gen-AI usage in music learning associated with both music-specific domain skills and dependence on AI tools?

RQ2: Do these two variables mediate the relationship between Gen-AI usage and performance quality?

RQ3: Does AI literacy moderate the relationships between Gen-AI usage and the two mediators?

This study makes both theoretical and practical contributions by offering an extended analysis of how generative AI (Gen-AI) is associated with cognitive and metacognitive processes in music learning. In doing so, it extends the application of SRL and CLT to AI-mediated creative education in three specific ways, suggesting that Gen-AI may function not only as a neutral instructional tool but as an active cognitive agent that is associated with changes in learners' self-regulation and cognitive processing. First, it illustrates how self-regulation in AI-rich settings may operate as co-regulation distributed between the learner and an intelligent system, such that planning, monitoring, and evaluation are partially redistributed to the AI rather than residing solely within the learner. Gen-AI does not create uniformity in learners' learning outcomes; instead Gen-AI appears to operate through two distinct psychological pathways: (1) as a scaffold to support the learner during skill acquisition and to create cognitive efficiency and (2) as a means to encourage cognitive offloading and reduce self-regulatory engagement, depending on how learners use the technology and their fluency with it. Second, it refines the application of CLT by showing that Gen-AI does not lower cognitive load uniformly for all learners; it may raise or lower germane load depending on the learner's AI literacy, with deeper schema construction and longer retention reflecting stronger germane processing. Third, it positions AI literacy not merely as a technical skill but as a higher-order metacognitive capability aligned with self-regulation. This perspective suggests that AI literacy is a key condition shaping whether Gen-AI functions as a helpful scaffold or a cognitive replacement. This two-way process provides a more nuanced psychological understanding of technology supported learning than the models of cognitive enhancement or cognitive impairment. In a practical sense, the study offers implications to both educators and curriculum designers by emphasizing the importance of using Gen-AI in a metacognitive way that promotes active usage and engagement rather than

dependence. The study suggests that music education should involve the development of AI literacy, with the aim of using Gen-AI as a cognitive scaffold and not as a substitute for the cognitive process of music-making. This study offers a timely psychological perspective on the interplay between human cognition and AI in education, focusing on learning, performance development, and creative agency in the current educational landscape.

2. Theoretical background

2.1. Self-regulated learning theory

Self-regulated learning (SRL), as conceptualized by Zimmerman (2002), denotes the ways in which learners actively monitor, control and reflect upon their cognitive, affective and behavioral processes in the pursuit of academic achievement. SRL is not a distinct trait, but rather an interactive cyclical process that proceeds through three phases of forethought (strategic planning, goal setting), performance (self-regulation, self-monitoring), and self-reflection (self-evaluation, adaption). In the domain of musical learning SRL is seen in activities such as goal-setting for practice, choice of repertoire, monitoring intonation or rhythm accuracy, and formulating modifications in light of feedback. The incorporation of Gen-AI entails the introduction of new phenomena in this process. On the one hand, AI can function as an external regulator, providing real-time feedback, making recommendations for modification, and registering improvement, thus facilitating learners' self-regulative behavior (Afzaal et al., 2021). For example, a difficulty-adjusting AI-based ear training app enables the performance stage of SRL through continuous self-monitoring by the user. In addition, SRL theory cautions against the use of displaced agency (Oppong et al., 2019). If students simply take for granted the output of AI algorithms without critically analyzing it, they may lose motivation to engage in metacognition and impair their internal regulatory mechanisms. This may result in a lack of self-efficacy and motivation, especially in the absence of AI or when it is not available, such as during live performance. Hence, while Gen-AI may support self-regulation when used strategically, it may undermine it when used non-strategically. SRL theory thus provides a compelling framework to understand the use of AI tools as a failure in self-regulation. Those students who possess sound self-regulatory capacities are more likely to use AI systems as educational tools rather than a dependence. In this regard, learners can view their AI literacy as a metacognitive resource that enables them to self-regulate while interacting with AI-based applications through effective goal-setting, self-monitoring, and self-evaluation, instead of AI directly determining their learning outcomes. Students who do not have such capacities will rely on it and become dependent, outsourcing mental work and even creative judgment to the machine.

2.2. Cognitive load theory

Cognitive Load Theory (Sweller, 2010) identifies how the design of instructional material can influence how learners learn through the cognitive load imposed on working memory. CLT identifies three types of cognitive load: intrinsic cognitive load, which refers to the inherent complexity of the content being learned (e.g., the difficulty of learning polyrhythms or complex harmonic progressions); extraneous cognitive load, which is the cognitive effort associated with the poor design of instructional materials (e.g., the use of complex interfaces or the inclusion of irrelevant or unnecessary information); and germane load, the effort invested in schema construction and deep learning (e.g., executing a piece slowly to absorb finger patterns) (Sweller, 2010).

Effective instructional material reduces extraneous cognitive load and maximizes germane cognitive load in order to promote the long-term development of learners' musical schemas (Hultberg et al., 2018). Gen-AI technologies can cap extraneous cognitive load on music learning. For example, AI-generated accompaniment allows

soloists to rehearse without the need for a pianist, thereby alleviating much of the coordination issues that often arise (White, 2025). Similarly, AI transcription tools can translate an audible sound into notation, thus allowing learners to use their mental resources to make interpretive decisions instead of dedicating them to decoding pitch and rhythm (Bhattarai & Lee, 2023). The overall goal of these approaches is to adhere to the principles of effective instructional design as outlined in Cognitive Load Theory by simplifying learning tasks and promoting effective practices related to collegiate music education. However, Gen-AI may incur an unnecessary load when poorly designed or misapplied. Complex AI user interfaces, ambiguous or vague feedback, or too much information from multiple AI suggestions can inundate working memory. Furthermore, one of the possible factors affecting germane load is cognitive offloading, which is the tendency to use external systems for cognitive work (Costley, 2020). Without knowledge of the rules, repeating patterns or corrections with the help of AI does not lead to the development of schemas and may have a negative effect on long-term memory. With regard to cognitive resources, based on CLT, Gen-AI can facilitate or impede learning. The implications of AI literacy are higher: Understanding when and how to use AI can make the difference between enhancing the learning process and hindering it. Thus, AI literacy operates as a boundary condition that shapes the association between Gen-AI usage and learning outcomes, rather than as a factor that directly amplifies or suppresses cognitive load mechanisms.

3. Hypotheses development

3.1. Gen-AI usage and music-specific domain skills

For music skills such as aural perception, music theory, technique, and improvisation, Gen-AI can serve as a scaffolding tool for learners. According to the self-regulated learning theory, purposeful learning involves self-regulation processes and adaptation, which can all be enhanced through the use of AI feedback (Carter Jr et al., 2020; Zimmerman & Schunk, 2012). For instance, AI-based ear training systems can adapt to the student's performance level, keeping them within their "sweet spot" of difficulty (Seufert et al., 2024). Likewise, AI-enabled computer programs that support composing music may generate harmonic sequences and voice-leading patterns that learners use as learning materials (Colafiglio et al., 2024). From the perspective of CLT, these tools reduce unnecessary cognitive load by computerizing repetitive tasks like notation input (Sankaranarayanan et al., 2024). Empirical evidence from technology-enhanced learning contexts shows that automation enables working memory to be employed in germane cognitive actions, i.e., schema construction and pattern detection (Mo et al., 2022). Hence, it is possible to expect that purposeful interaction with Gen-AI within the music learning process will be associated with stronger music-specific domain capability by means of cognitive effectiveness and metacognitive support. Hence, we predict that:

H1a. : Gen-AI usage in music learning is positively associated with music-specific domain skills.

3.2. Gen-AI usage and dependence on musical AI tools

Gen-AI technology, although facilitating learning, poses the danger of leading to cognitive offloading due to its ease of use and constant delivery of high-level results. Cognitive offloading refers to the tendency to rely on external systems to perform intellectual tasks that would normally be performed internally (Risko & Gilbert, 2016). Regular application of AI in the music learning process in activities such as melodic composition and rhythmic correction may encourage dependence behavior. As defined in the Introduction, this dependence reflects maladaptive cognitive offloading rather than mere frequency of use. The music education role of AI is complex, offering both generative (creativity support) and ancillary (support for music learning) roles

(Atanacković, 2024). While AI has the potential to increase the quality of learning experiences, motivation, engagement, creativity, and learning accomplishment (L. Chen, 2025), it also presents potential pitfalls. The largest area of concern is over-reliance on AI-based software, which may impede the development of basic musical skills and self-regulating functions (Macnamara et al., 2024). From the perspective of SRL, this over-reliance would indicate failure in the forethought and self-monitoring phases. Students may fail to set personal goals of improvement or fail to reflect on their work when no feedback from the AI is supplied (Cloude et al., 2022). CLT also explains this: as AI continuously reduces extraneous and intrinsic load, students may become so accustomed to low-effort processing that they bypass the germane load necessary for mastery internalization (Reynaert et al., 2026; Seufert et al., 2024). This cycle fosters dependency, as students rely on AI to make things easier, thereby reducing their capacity for self-reliant work. Convenience is associated with decreased self-efficacy and skill decay in other domains such as calculator use in math and grammar checkers in writing (Lin et al., 2024). Therefore, more use of Gen-AI is expected to be associated with heightened dependency, particularly without structured guidance. Therefore, we predict that:

H1b. : Gen-AI usage in music learning is positively associated with dependence on musical AI tools.

3.3. Music-specific domain skills and performance quality

Technique, feeling for the music, and timing are important aspects of music performance. In the context of SRL theory, if the learners get better at tasks such as sight-reading and playing by ear, then they are able to self-regulate their learning process to become better learners (Atabay & Çakıroğlu, 2025). This establishes the cognitive resources needed to maintain high-level performance even in stressful situations (Dörrenbächer-Ulrich & Bregulla, 2024). Learners with higher levels of domain skills will then have more opportunities to achieve higher levels of technical accuracy, expressive interpretation, and confidence in performance, which together will define the quality of the performance. This is supported by CLT, stating that automation of basic skills such as finger patterns and chord forms, reduces the cognitive load that is intrinsic to the cognition of the musician, allowing them to concentrate on phrasing and dynamics (Goopy, 2026; Leppink, 2020). Studies highlight that mastering these fundamentals through deliberate practice is strongly associated with performance outcomes. For instance, highly skilled musicians demonstrate superior intonation and ensemble coordination (Wolf & Kopiez, 2018), and those with strong theoretical backgrounds improvise more fluently (van Vreden, 2021). Therefore, the development of music-specific domain skills is closely associated with performance quality. Hence, we propose that:

H2a. : Music-specific domain skills are positively associated with performance quality.

3.4. Dependence on musical AI tools and performance quality

Over-dependency on AI technology may detract from the formation of autonomous musical agency, an essential aspect of high-caliber performance. With increasing dependence on AI to compose, improvise, or correct mistakes, learners may deprive themselves of fundamental cognitive activities for schema construction and procedural memory strengthening, crucial aspects in CLT (Sankaranarayanan et al., 2024; Valcke, 2002). This reliance brings difficulty in high-stakes or unstructured tasks where there is no support from AI, such as live performances or auditions (He & Chen, 2024). From an SRL perspective, this lack of self-regulation reflects reduced engagement in self-assessment, goal setting, and strategy application by learners in their musical development (Martin, 2004). This is in line with Zimmerman's SRL model, in which the repetitive cycles of forethought, performance, and self-reflection are prominent in music (Dörrenbächer-Ulrich & Bregulla,

2024). The weakening of these SRL-related processes may inhibit the acquisition and long-term retention of skills (Wang et al., 2023). Consequently, this impairment may undermine not only technical proficiency but also expressive control and performance confidence, as learners struggle to adapt, interpret, and perform effectively without AI support. In addition, dependency on AI may be associated with reduced self-efficacy, which could increase anxiety and lower motivation when working individually (Bandura, 1977; Mieder & Bugos, 2017). Self-efficacy, as defined by Bandura, is the belief in one's ability to succeed in specific situations (Bandura, 1977). When learners receive continuous solutions to their musical tasks via AI, there is a possible inhibition in the development of their own problem-solving resilience and ability to cope with failure in music (Cooke et al., 2026; Orejudo et al., 2021). Educational technology research supports the view that over-reliance on automated tools is associated with less long-term retention and poorer transferability of learned information (McCrudden et al., 2010). This may include an inability of the music learner to compose without AI prompts, or to fix errors in real time (Zeng et al., 2023). Conversely, students with high self-efficacy are more likely to engage in effective SRL strategies, such as help-seeking and effort regulation (Anthonysamy et al., 2020). Therefore, while over-reliance on AI may optimize short-term output, it may diminish the complexity and permanence of musical performance. Hence, we propose that:

H2b. : Dependence on musical AI tools is negatively associated with performance quality.

3.5. Music-specific domain skills mediation effects

The integration of Gen-AI into musical education is expected to be associated with the quality of performance through enhanced music-specific skills. This aligns with SRL and CLT. SRL assumes that learners who intentionally use AI to provide comments, imitate accompaniment, or investigate composition practice are engaging in a consciously goal-oriented manner, which increases self-regulation and mastery of skills (Shi et al., 2025; Xu et al., 2025). This is associated with better music-specific skills and improved quality of performance. CLT assumes that AI technologies reduce extraneous cognitive load through automation of processes like notation, allowing the learner to assign cognitive resources to expressive or technical interpretation (Muthmainnah et al., 2024). This process allows for the formation of robust schemas and better quality of performance (A. Chen et al., 2025). Personalized learning technologies made possible by AI facilitate adapted learning paths and development of cognitive ability (Kumaresan et al., 2025). Music-specific skills are therefore expected to mediate the relationship between Gen-AI usage and improved performance, especially with reflective and intentional practice. Therefore we hypothesize:

H3a. : Music-specific domain skills mediate the association between Gen-AI usage and performance quality.

3.6. Dependence on musical AI tools mediation effects

Generative AI usage has the potential to be negatively associated with performance indirectly to the extent that learners are overly reliant on AI tools (Karny et al., 2024). The current study indicates that this negative mediation may represent a paradox of ease. While AI augments the ease of task completion in the short term, greater reliance appears to inhibit the ability of the learner to complete tasks independent of AI. From an SRL perspective, reliance is disruptive of the self-regulatory cycle in that the learner who relies on AI may cease to set personal goals, monitor goal progress, or adapt strategies to be used independently of AI (Jin et al., 2023). This may lead to passive learning and a reduced perception of skill ownership. CLT explains this phenomenon more specifically in that reliance on AI offloads the cognitive load of task performance to the AI; thus, the user eludes the germane load required

for schema construction, and the knowledge gained may be of a fragile, context-dependent nature (T. Wang et al., 2023). As a result, performance may suffer in AI-free settings. This pattern mirrors findings in other domains, such as students relying on grammar checkers who perform worse in unassisted writing tasks (Liu et al., 2024). Similarly, dependence on AI tools has been shown to potentially diminish higher-order thinking skills (Y. Wang & Liu, 2024). Thus, dependence may act as a hidden cost of Gen-AI usage, mediating a negative indirect association with performance quality despite the tool's surface-level benefits. Hence, we propose that:

H3b. : Dependence on musical AI tools mediates the association between Gen-AI usage and performance quality.

3.7. AI literacy moderation

To fully harness the positive link between Gen-AI and skill acquisition, AI literacy — which encompasses understanding, critically assessing, and responsibly applying AI technologies — is essential (Ng et al., 2021). In this study, AI literacy is not considered a direct antecedent of performance but rather a metacognitive self-regulatory ability that influences the translation of AI usage into learning processes. As distinguished in the Introduction, AI literacy extends beyond general digital literacy by encompassing the capacity to evaluate generative outputs critically and to judge when delegating a task to AI supports rather than replaces learning. Learners who have high AI literacy apply AI tools strategically, knowing their affordances as well as limitations (Dalgıç et al., 2024). These students are effectively applying SRL strategies through setting clear learning goals, and critically evaluating the outputs from the AI, rather than simply accepting them. This is because reflective use strengthens engagement and internalization of skills (Korte et al., 2024). From the CLT perspective, AI-literate learners are better able to manage cognitive load by reducing extraneous demands while preserving germane load for active learning (Sweller, 2010). Conversely, learners with low AI literacy may misuse AI or fail to use it in constructive ways. AI literacy is therefore an enabling condition that strengthens the pathway from Gen-AI use to skill development, resulting in better learning outcomes across a variety of domains (Dalgıç et al., 2024; Zhao et al., 2022). Empirical studies show that digital literacy moderates the effectiveness of technology in education (Sumin et al., 2021). Therefore, we propose that:

H4a. : AI literacy strengthens the positive association between Gen-AI usage and music-specific domain skills.

AI literacy acts as a shield against over-reliance on AI technologies through a deeper appreciation of AI operation and limitations (Almatrafi et al., 2024; Chiu et al., 2024). Accordingly, AI literacy functions as a boundary condition that shapes the extent to which Gen-AI usage translates into cognitive offloading tendencies, rather than acting as a variable that independently drives dependence. More AI-literate individuals are more likely to critically review AI outputs, realizing that AI content is probabilistic and that bias can occur. This reduces the likelihood of blindly taking on AI-generated content, thereby enabling the more judicious use of AI (Bewersdorff et al., 2025). From the SRL point of view, the knowledge of AI promotes students' autonomy and self-efficacy, allowing them to use AI tools not just as tools, but as partners. AI helps students practice effective SRL strategies, including setting objectives, tracking progress, and adjusting study methods, which aids in achievement of their learning goals (Du et al., 2025). CLT proposes that AI-literate learners are more discerning in deciding when and how to offload to AI, thus preventing unnecessary cognitive offloading, which impairs schema building (Järvelä et al., 2023). With their awareness of the fundamental mechanisms of AI, learners can maximize its application to enhance cognition without sacrificing learning. For example, a student who is aware that AI generates work using existing patterns can use it for idea generation, but will still apply critical

thinking and individual composition to ensure originality and deeper understanding (Bécue et al., 2024). AI ethics and education research clarifies how AI literacy mitigates blind trust in algorithmic systems (Williams, 2024). Thus, AI literacy is expected to attenuate the tendency to become dependent on AI, even with frequent usage, making it a crucial moderator in the responsible integration of AI in music learning. Hence, we propose that:

H4b. : AI literacy weakens the positive association between Gen-AI usage and dependence on musical AI tools.

Fig. 1 represents the research model.

4. Methodology

4.1. Data collection

The study was conducted in six of China's top music schools, namely the Central Conservatory of Music (Beijing), Shanghai Conservatory of Music, China Conservatory of Music, Sichuan Conservatory of Music, Xinghai Conservatory of Music (Guangzhou), and Wuhan Conservatory of Music. These schools were selected due to their national reputation, multilingual and multinational student body, and dedication to integrating digital tools into instruction and practice. The changing technology environment in China for educational purposes, as it is being shaped by governmental plans for future development such as the "Education Informatization 2.0 Action Plan," provides a meaningful environment to examine the integration of AI into creative disciplines. A stratified convenience sample was used to achieve representation by academic level (undergraduate, graduate), instrument group, and genre. The target population was individuals 18 years of age and older who had received at least one year of formal training in music and had been involved in using some digital environment in music, such as digital audio workstations (DAWs), notation programs, or web-based practice programs. Participation was voluntary and informed consent was obtained before survey access. The sample finally utilized was 1142 usable cases after elimination of those who had not completed answers and/or whose response patterns on the survey showed straight-lining. The sample comprised 56% female and 44% male and had a mean age of 20.8 years ($SD = 1.9$). Sixty-one percent were undergraduates, while 39% were in post-graduate programs. Instrumental distribution: piano (30%), strings (25%), voice (19%), Chinese traditional instruments (14%), woodwinds/brass (9%), and composition (3%). Length of formal music study experience averaged 3.1 years ($SD = 1.4$). A power analysis with G*Power showed that the sample size is more than adequate to detect a medium effect ($f^2 = 0.15$) in multiple regression given eight predictors, $\alpha = 0.05$, and power = 0.95. The sample is also highly representative within the target population and has high statistical power with clearly more than ten observations per estimated parameter, and good parameter estimates.

For the qualitative phase, the data were gathered from 15 field experts, with experience of the use of digital and AI tools in music learning and music performance contexts. A purposive sampling approach was used to ensure that participants had the opportunity to share their experience and expertise in the areas of interest, as well as their personal experiences with the use of generative AI technologies in the musical domain. This research utilized semi-structured interviews to allow for greater exploration of the expert participants' perspectives on generative AI's association with skill acquisition, dependency, quality of performance, and developing AI literacy. Each interview took approximately 30–45 min and was conducted using remote web conferencing to accommodate participants' availability and geographic locations. Prior to the interviews, participants were provided with an explanation of the purpose of the study, and they signed an informed consent form allowing for participation. The sample size of 15 experts was deemed appropriate for qualitative inquiry, as it allowed for sufficient depth of insight while achieving thematic saturation, where no substantially new

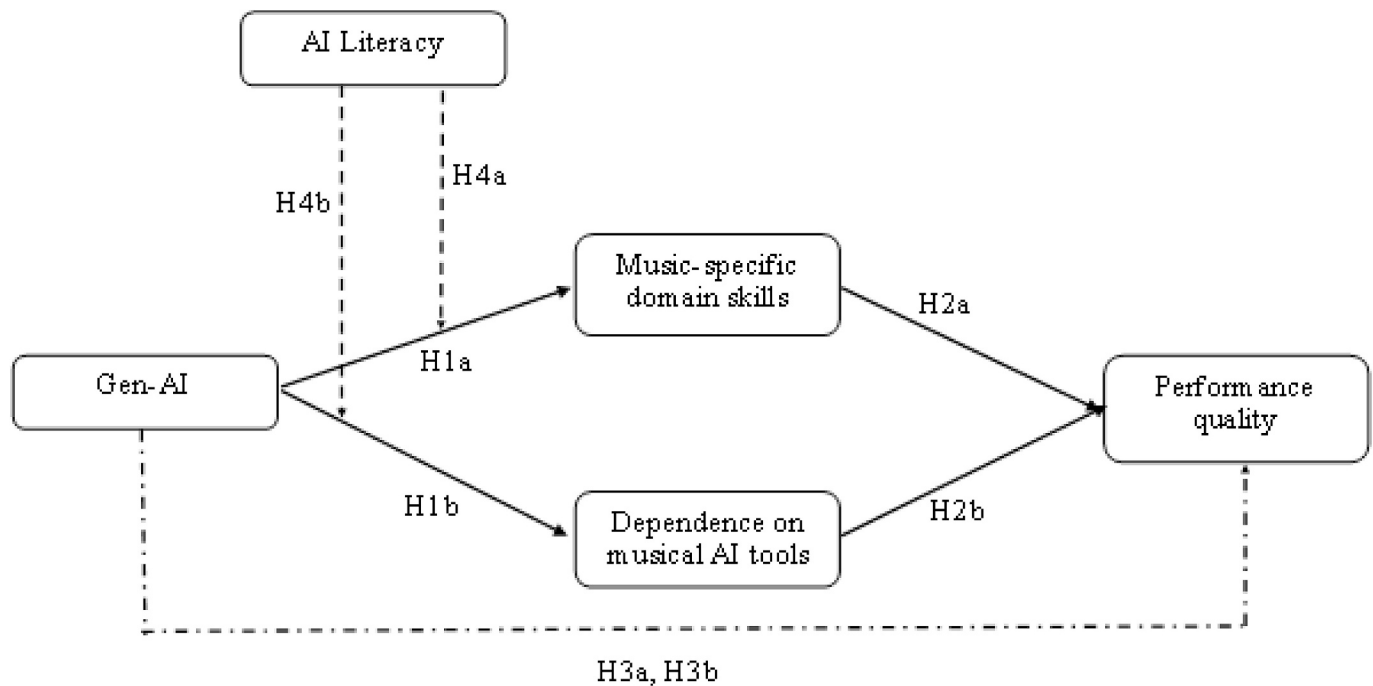


Fig. 1. Research model.

themes emerged from additional data.

4.2. Measures

All constructs were measured using multi-item, self-report scales adapted from validated instruments in educational technology, music psychology, and AI literacy research. A 7-point Likert scale (1 = Strongly Disagree to 7 = Strongly Agree) was used for all items. Gen-AI refers to AI systems capable of generating novel musical content such as melodies, harmonies, rhythms, or full compositions, based on user inputs or prompts. Unlike rule-based systems, Gen-AI models (e.g., transformer networks) learn from vast datasets of musical works and can produce stylistically coherent outputs (Wei et al., 2025). Gen-AI usage in music learning was assessed with nine items adapted from Khan et al. (2025) and S. Wang and Zhang (2025). The study refers to generative AI tools as AI-based systems commonly used in music learning for tasks such as composition, accompaniment generation, practice support, and musical idea development; these include ChatGPT and other AI-enabled music generation or assistance platforms available to students. Music-specific domain skills refer to the core competencies required for musical proficiency, including aural skills (pitch, rhythm, harmony), music theory, technical facility, sight-reading, and expressive interpretation (Wei et al., 2025). Music-specific domain skills were measured using five items assessing aural perception, music theory knowledge, technical fluency, sight-reading ability, and improvisational skill, adapted from Mingjie et al. (2025) and Ripani (2026). Dependence is defined as a psychological and behavioral reliance on AI tools to perform tasks that learners could otherwise accomplish independently (Zhai et al., 2024); in this study it is operationalized specifically as a maladaptive form of cognitive offloading that reflects reduced engagement in self-regulated learning processes. We gauged dependence on musical AI tools using three items adapted from the cognitive offloading and technology dependence literature, capturing the extent to which participants rely on AI tools to complete musical learning and performance-related tasks (Mingjie et al., 2025). Our use of just three questions focused specifically on the maladaptive kind of cognitive offloading, not a broader range of possibilities. Short, three-item scales like the current study are actually typical for targeted reflective constructs in SEM research. The scale

showed good reliability and convergent validity in our sample, with Cronbach's α and composite reliability (CR) both at 0.81 and average variance extracted (AVE) at 0.597, all above recommended thresholds. The AI literacy measure included five items adapted from Ji et al. (2025), assessing participants' understanding of generative AI, their ability to critically evaluate AI outputs, and their awareness of ethical considerations. The quality of performance is evaluated based on the musical performance itself (technical, expressive, fluent, and creative aspects) (Gonzalez-Sanchez et al., 2019). The current research examines the quality of the performance in terms of multiple facets such as technical skill (accuracy, fluency), interpretation, and confidence in performance. The study relied on five self-reported measures of technical accuracy, expressiveness, fluency, and confidence to measure this. Since quality of performance was subjectively assessed by the learners and not independently by the assessors, it is referred to as perceived performance quality throughout and its implications are discussed in the limitations section.

Items were translated by two bilingual music ed. researchers. The items were pilot-tested with 45 students from three schools to ensure clarity, cultural appropriateness, and response consistency. The changes were made to reflect the feedback, such as reducing tech jargon and modifying some of the idioms. Following the pilot, the full survey was administered through Wenjuanxing, a widely used Chinese online survey platform. We shared it through WeChat groups created by the faculty coordinators and through faculty email lists. The data was collected from April to June 2025. We took into account tips from Podsakoff et al. (2003) in an effort to minimize common method bias during design. Participation was entirely anonymous and voluntary and we made sure respondent anonymity in order to reduce their worry and pressure to respond dishonestly. Moreover, predictor and criterion items are divided into separate sections of a questionnaire. During the piloting process we also made some wording changes to reduce ambiguity. Participants gave electronic consent and assured that the responses were anonymous before they completed the online survey. The survey did not collect any personally identifying information. Furthermore, all the data was kept on secured servers that had encryption and could only be accessed by the core research team.

4.3. Data analysis

Data were analyzed through Covariance-Based Structural Equation Modeling (CB-SEM), which is well suited to confirmatory testing of theoretically specified models involving mediation and moderation (Dash & Paul, 2021). This process was done in a sequence. First, the measurement model was assessed by performing confirmatory factor analysis (CFA), analyzing the reliability of the instrument, Cronbach's alpha, composite reliability, convergent validity (AVE > 0.50), and discriminant validity (Fornell & Larcker, 1981 and the HTMT method). Then we estimated the structural model which comprised all the direct, indirect, and moderated paths. To figure out moderation effects, we made latent interaction terms using the unconstrained approach from Marsh et al., 2004. This included developing product indicators for AI literacy and Gen-AI—the latter was used as a latent interaction term to predict the mediators. Model fit was examined with multiple indices: $\chi^2/\text{df} < 3$, CFI > 0.95, TLI > 0.95, RMSEA < 0.06 (90% CI upper bound < 0.08), and SRMR < 0.08. Indirect effects (H3a and H3b) were tested using bootstrapping (5000 resamples) to obtain bias-corrected confidence intervals. The large sample size and the use of a robust maximum likelihood estimation method give the model high inferential power for testing the hypothesized relationships under a confirmatory framework.

Expert interviews provided qualitative data and were analyzed using thematic analysis to provide valuable insights. The interviews were transcribed and examined multiple times to ensure the experts were familiar with the data. Open coding was employed to generate initial codes relating to Gen-AI use, skill development, dependence, performance quality, and AI literacy. These codes were then grouped into higher-order categories through axial coding, yielding the key themes aligned with the study's conceptual framework. To enhance rigor, a second coder reviewed a random sample of transcripts. Any discrepancies were discussed to ensure consistency and reliability. Last, the themes were examined with regard to the quantitative results in order to deepen our understanding of the psychological processes and to support the integration of the results in a mixed-methods approach.

5. Results and discussion

5.1. Preliminary analysis

To ensure that the requirements of CB-SEM are fulfilled, data were first checked for normality, reliability, and common method bias. As shown in Table 1, the standard deviations of all constructs were within the good variability range (from 0.83 to 0.91), indicating sufficient variation in data. The overall mean scores of each of the constructs were near the middle of the Likert Scale range ($M = 3.98\text{--}4.12$), indicating a generally neutral-to-moderately positive self-evaluation. CB-SEM requires normality, particularly in the case of maximum likelihood estimation. We checked skewness and kurtosis to figure out the distribution of composite scores for each construct. Skewness ranged from -0.18 to 0.11 , well within the ± 1.0 range, indicating no problematic asymmetry. Kurtosis values ranged from -0.23 to 0.21 , indicating no meaningful deviation from a mesokurtic distribution. If the absolute skewness and the absolute kurtosis are less than 2 and 7, respectively, then ML estimation can be used. As the observed values were close to zero, the assumption of normality was well supported. Thus, the assumption of

Table 1
Descriptive statistics.

Construct	Mean	Std. dev	Skewness	Kurtosis
GenAI	4.12	0.89	−0.18	0.21
Domain skills	4.05	0.83	−0.12	−0.09
Dependence	3.98	0.91	0.05	−0.23
AI literacy	4.01	0.85	0.11	−0.15
Performance	4.07	0.87	−0.09	−0.18

multivariate normality in CB-SEM is satisfied.

All self-reported data were assessed for common method variance using Harman's single factor test suggested by Podsakoff et al., 2003. Thus, all 27 items were examined in an unrotated exploratory factor analysis. The first factor accounted for only 31.2% of the variance, which is below 50%. Thus, there's no dominant single latent factor in the data. However, this single factor test is rather insensitive, so that we cannot exclude common method bias with this test. It is therefore treated as one of several safeguards alongside the procedural remedies, and the possibility of residual method variance is revisited in the limitations.

5.2. Confirmatory factor analysis

We used confirmatory factor analysis to check our model's fit before estimating structural paths. A number of fit indices were examined in Table 2. The Comparative Fit Index (CFI) is 0.943, and the Tucker-Lewis Index (TLI) is 0.929, which are both above the suggested level of 0.90 for a good fit (Hu & Bentler, 1998). Hu and Bentler (1998) suggest using a cutoff value of 0.08 for SRMR and 0.06 for RMSEA; values of RMSEA less than 0.08 indicate that the model is an acceptable fit. The root mean square error of approximation (RMSEA) = 0.007 is small, demonstrating close fit. The χ^2/df ratio of 1.061 is well below the upper limit of 5.0. Taken together, these indices indicate that the measurement model fits the data well. This validates the latent structure and justifies moving on to test the structural model.

5.3. Measurement model assessment

We did a confirmatory factor analysis with CB-SEM to make sure our latent constructs were valid and reliable before testing the structural model. The measurement model was evaluated for convergent validity, reliability, and discriminant validity. All analyses were performed using CB-SEM with maximum likelihood estimation in AMOS.

To check convergent validity, we used three tests: factor loadings at least 0.70, Composite Reliability of 0.70 or higher, and Average Variance Extracted of at least 0.50 (Hair et al., 2019). According to Table 3 and Fig. 2, every item fit well into its construct with loadings between 0.63 and 0.84. Plus, all constructs showed strong reliability (CR of 0.81 to 0.922) and good-to-strong AVE (0.51 to 0.657), proving the convergent validity.

Discriminant validity ensures that each construct is empirically distinct from the others. It was assessed using two complementary methods: the Fornell and Larcker (1981) criterion and the HTMT (Heterotrait–Monotrait) ratio. The square root of AVE for each construct should be greater than its correlations with other constructs; per the Fornell and Larcker (1981) criterion, discriminant validity is acceptable. HTMT values < 0.85 (or < 0.90) indicate sufficient discriminant validity (Henseler et al., 2015); per HTMT, discriminant validity is also acceptable (Table 4).

Table 2
Goodness-of-fit values.

Fit indices	Recommended value	Source	Estimated model
Chi-square		(Hu & Bentler, 1998)	333.247
P value	In-significant		0
ChiSqr/df	≤ 5		1.061
RMSEA	≤ 0.06		0.007
GFI	≥ 0.90		0.952
SRMR	≤ 0.08		0.017
NFI	≥ 0.90		0.954
TLI	≥ 0.90		0.929
CFI	≥ 0.90		0.943

Note: A significant χ^2 is expected given the large sample size ($N = 1142$), as the χ^2 statistic is highly sensitive to sample size; alternative fit indices are therefore emphasized.

Table 3
Convergent validity and reliability.

Construct	Item	Loading	VIF	α	CR	AVE
GenAI	GenAI_1	0.7	1.846	0.922	0.922	0.569
	GenAI_2	0.684	1.796			
	GenAI_3	0.781	2.282			
	GenAI_4	0.771	2.211			
	GenAI_5	0.808	2.468			
	GenAI_6	0.693	1.811			
	GenAI_7	0.708	1.849			
	GenAI_8	0.841	2.85			
	GenAI_9	0.787	2.331			
Domain Skills	DS_1	0.633	1.492	0.836	0.837	0.51
	DS_2	0.776	1.954			
	DS_3	0.773	1.926			
	DS_4	0.657	1.541			
	DS_5	0.721	1.767			
Dependence	Dep_1	0.645	1.527	0.81	0.81	0.597
	Dep_2	0.811	1.996			
	Dep_3	0.846	2.076			
AI_Literacy	AIL_1	0.81	2.418	0.905	0.905	0.657
	AIL_2	0.775	2.167			
	AIL_3	0.852	2.787			
	AIL_4	0.857	2.836			
	AIL_5	0.754	2.053			
Performance	Perf_1	0.799	2.285	0.895	0.896	0.634
	Perf_2	0.797	2.296			
	Perf_3	0.86	2.822			
	Perf_4	0.695	1.761			
	Perf_5	0.822	2.458			

5.4. Structural model

The structural model was estimated using Covariance-Based Structural Equation Modeling (CB-SEM). Overall model fit was excellent, as corroborated by several fit indices: CFI = 0.965 and TLI = 0.962, both well above the 0.90 cutoff, indicating strong model–data fit. RMSEA was 0.039 with a 90% confidence interval of [0.036, 0.041], well below the 0.06 cutoff, indicating close approximate fit. SRMR was 0.075, within the recommended limit of 0.08. With a χ^2/df ratio of 2.99, the model demonstrates a high degree of parsimony and empirical adequacy for testing the hypothesized relationships.

All direct associations were statistically significant, as indicated in Table 5 and Fig. 3. GenAI usage was positively and significantly associated with music-specific domain skills ($\beta = 0.516$, $p < .001$), supporting H1a and showing that students who use generative AI tools more actively report greater improvement in aural perception, technical fluency, and music theory. Analogously, dependence on AI tools ($\beta = 0.542$, $p < .001$) was found to be significantly and positively related to GenAI usage, supporting H1b. The associations of the mediators with performance quality were also significant. As expected, music-specific domain skills were strongly related to performance quality ($\beta = 0.94$, $p < .001$), supporting H2a and reinforcing the salience of skill mastery in musical achievement. The magnitude of this coefficient should nevertheless be interpreted with caution, as both constructs were measured through same-source self-perceptions, which may inflate their shared variance. On the other hand, the use of AI tools had a significantly negative correlation with the quality of performance ($\beta = -0.655$, $p < .001$). This provides evidence for H2b and suggests that using AI tools more may be related to lower performance, which is in line with lower self-efficacy and less ability to solve problems on one's own. The R^2 values for music-specific domain skills, AI tool reliance, and performance quality are 0.437, 0.552, and 0.537, respectively, showing substantial explanatory power. Additionally, the interaction between GenAI usage and AI literacy was positively associated with music-specific domain skills ($\beta = 0.175$, $p < .001$). Thus, H4a is accepted and it suggests that the benefits of GenAI are greater for learners with higher AI literacy. This is consistent with the SRL theory, as it predicts that the AI-savvy learners will reflectively use AI, exhibit goal-directed learning,

and apply feedback in a smart way. Conversely, the interaction term was negatively associated with dependence on AI tools ($\beta = -0.169$, $p < .001$), supporting H4b and suggesting that AI literacy attenuates the tendency toward over-reliance. The moderating effects suggest that the use of AI is dependent on literacy but also serves as a means to control AI's reliance by learners. Indirect association between GenAI and performance through domain skills was positive and significant (0.485, $p < .001$), thus supporting H3a. As seen in the indirect association through dependence, the results were negative and significant with H3b being supported ($p < .001$, -0.355).

5.5. Qualitative results

The results of the qualitative analyses identified a number of themes that capture the relationship between Gen-AI and music learning processes and outcomes. The data collected included findings from music education experts: professors at universities, professional musicians, and AI-assisted music practitioners. The findings are structured under four themes: (1) Gen-AI as a skill development facilitator, (2) Gen-AI as a source of cognitive offloading and dependence, (3) performance quality as a function of internalized competence and (4) AI literacy as a regulatory capability.

Theme 1: Gen-AI as a Facilitator of Music-Specific Skill Development. Throughout the report, the experts noted the importance of the enhancement of music-specific domain skills which can be realized through the use of the tools of Gen-AI, offering real-time feedback, structured guidance, and opportunities for creative exploration. As one expert said: “AI tools can help students understand harmony, rhythm, and structure much faster because they get immediate feedback and examples.” Another explained: “Students use AI to experiment with musical ideas, which supports their creativity and technical understanding.” These insights support the quantitative findings that Gen-AI usage is positively associated with music-specific domain skills, particularly through guided practice and increased cognitive engagement.

Theme 2: Gen-AI Encourages Dependence through Cognitive Offloading. Experts also noted that increased dependence is related to the use of Gen-AI, especially when students are depending on AI-generated outputs rather than their own cognitive and creative processes. One participant said: “Some students rely too much on AI to generate melodies instead of developing their own musical thinking.” Another commented: “When students depend on AI for corrections or composition, they become less actively involved in the learning process.” The qualitative findings are consistent with the quantitative findings, indicating that use of Gen-AI is positively correlated with reliance on AI tools, which indicates dependency or maladaptive cognitive offloading and decreased self-regulation.

Theme 3: Performance Quality Depends on Internalized Musical Competence. It was noted that the quality of performances is closely related to the extent to which the musical skills are internalized and not externally sustained. A master explained, “Strong performers are those who have internalized technique and musical understanding, not those who rely on external tools.” One said, “In live performance situations, students cannot depend on AI, so their true competence becomes visible.” These insights corroborate the quantitative findings, that music domain-specific skills are positively related to performance quality while dependence is negatively related to performance outcomes.

Theme 4: AI Literacy as a Regulatory and Metacognitive Capability. AI literacy emerged as a critical factor influencing students' engagement with Gen-AI tools and the differences among students. One participant commented: “Students who understand how AI works use it more critically and effectively, rather than blindly following its suggestions.” Another explained: “AI literacy helps students decide when to use AI and when to rely on their own skills.” These findings support the quantitative findings, suggesting that AI literacy further bolsters the positive relationship between Gen-AI usage and domain skills, and reduces the relationship between Gen-AI usage and dependence, making AI literacy an important

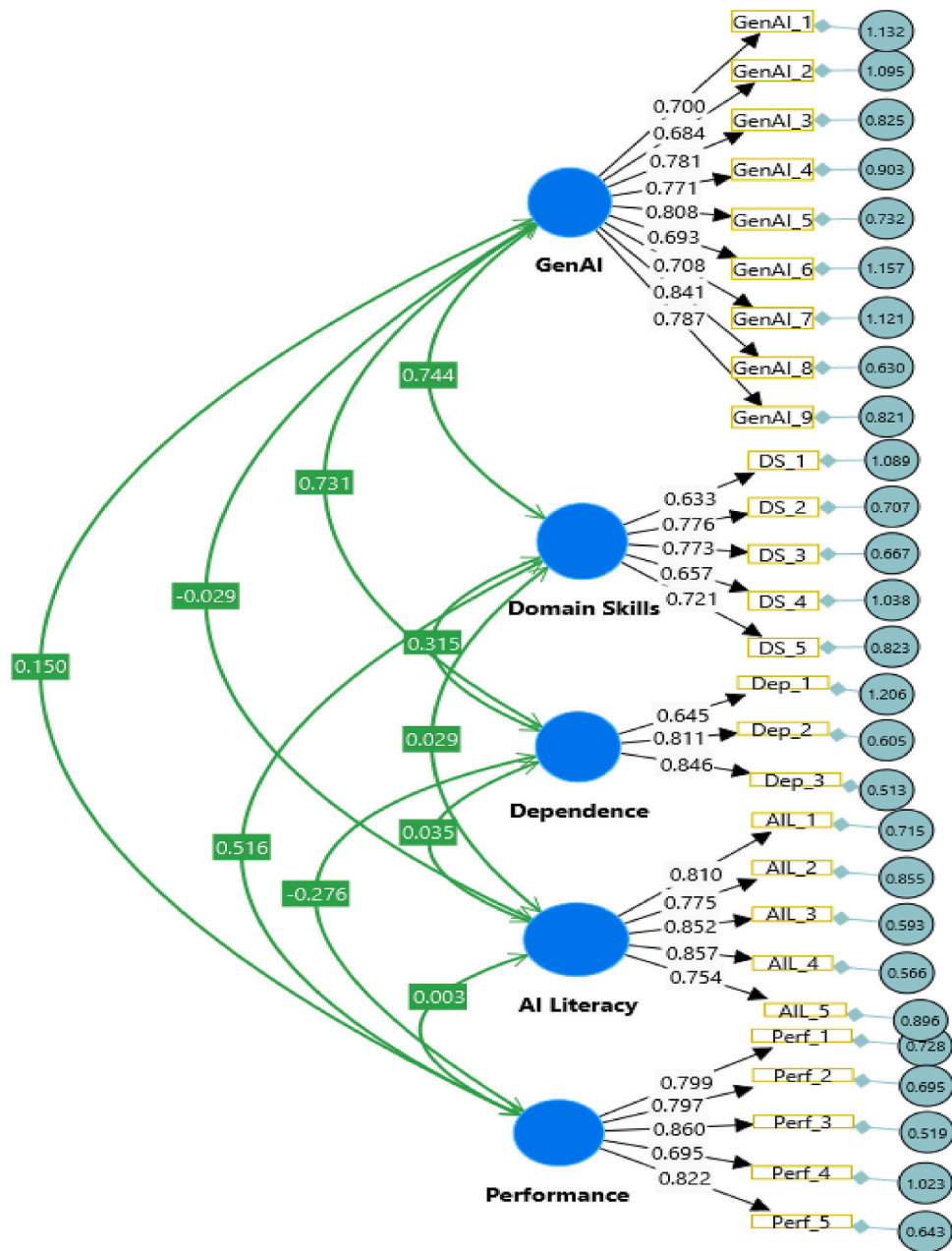


Fig. 2. Confirmatory factor analysis.

Table 4
Discriminant validity.

Construct	1	2	3	4	5
1. AI Literacy	0.811				
2. Dependence	0.035	0.773			
3. Domain Skills	0.029	0.315	0.714		
4. GenAI	−0.029	0.731	0.744	0.755	
5. Performance	0.003	−0.276	0.516	0.15	0.796

HTMT

1. AI Literacy					
2. Dependence	0.033				
3. Domain Skills	0.033	0.32			
4. GenAI	0.036	0.729	0.747		
5. Performance	0.026	0.275	0.518	0.153	

moderating variable.

Table 5
Hypotheses results.

Path	β	STDEV	T value	P values	Supported
GenAI - > Domain Skills	0.516	0.029	17.609	0.000	Yes
GenAI - > Dependence	0.542	0.031	17.689	0.000	Yes
Dependence - > Performance	−0.655	0.04	16.27	0.000	Yes
Domain Skills - > Performance	0.94	0.05	18.952	0.000	Yes
GenAI - > Domain Skills - > Performance	0.485	0.031	15.817	0.000	Yes
GenAI - > Dependence - > Performance	−0.355	0.025	13.959	0.000	Yes
InterAIL*DS - > Domain Skills	0.175	0.018	9.852	0.000	Yes
InterAIL*Dep - > Dependence	−0.169	0.018	9.207	0.000	Yes

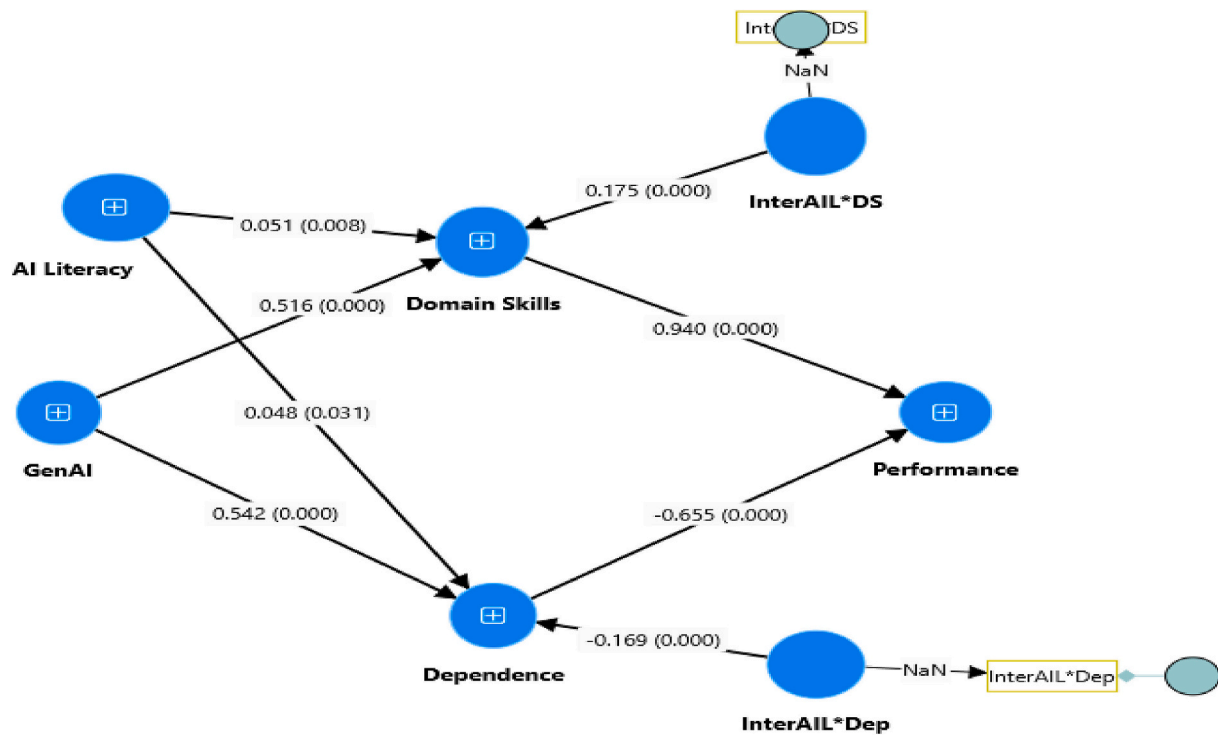


Fig. 3. Structural model.

5.6. Integration of quantitative and qualitative findings

A mixed-method design was applied in this research to gain an in-depth understanding of the relationships between Gen-AI use, music learning, and music performance quality. The qualitative results complement and extend the quantitative results. Experts further emphasize the positive link between Gen-AI use and the development of music domain skills, citing the role of AI in providing feedback, organizing practice, and fostering creativity. Second, this positive correlation between Gen-AI use and dependence on AI tools is mediated by experts' observations regarding cognitive offloading and decreased engagement in SRL. Thirdly, the contrasting linkages between domain skills and dependence and performance quality are corroborated by the focus on internalized competence as the basis for effective musical performance. Last, expert views that regard AI literacy as a metacognitive and regulatory process that facilitates strategic and reflective use of AI provide strong support for the moderating role of AI literacy. Overall, the integration of quantitative and qualitative findings shown in Table 6 demonstrates strong convergence, enhancing the validity and robustness of

the study.

5.7. Discussion

Because the present design is cross-sectional, the relationships reported below are interpreted as associations rather than causal effects, and the language of the discussion is framed accordingly. The advent of Gen-AI has brought new opportunities for personalized learning, creativity exploration and skill development in music education. However, relying on AI to carry out cognitive and creative tasks also poses risks of dependency, reduced self-confidence, and the lack of transferable music-specific skills when learners become reliant on AI. To address these gaps, this study designed and tested a moderated-mediation model linking Gen-AI usage to performance quality indirectly through music-specific domain skills and dependence on AI tools, with AI literacy as a moderator of the first-stage associations. This study, which draws upon SRL and CLT as theoretical underpinnings and applies these to a large sample of music students in University of China settings, offers a multi-faceted approach for understanding the complexities of music education integration of human learners and generative AI.

The results show that Gen-AI usage is significantly associated with music-specific domain skills and with dependence on AI tools, supporting the dual role that the use of Gen-AI plays in music learning. This confirms previous findings in the educational technology domain, which have indicated that intelligent tools can enhance learning and induce cognitive offloading (Mingjie et al., 2025). The positive correlations between frequency of using Gen-AI and domain proficiency align with the self-regulated learning theory of Zimmerman (2002), which suggests that self-regulated learning occurs when students make goals, self-monitor progress, and modify learning strategies, all of which could be facilitated with the use of Gen-AI in terms of feedback, adaptive exercises, and modeling of musical structures. However, the positive correlation with dependence indicates an increasing trend of cognitive offloading, where students are increasingly turning to AI to create melodies, correct rhythms, or provide stylistic options. In line with the framing, this reliance is interpreted as a negative pattern of cognitive

Table 6

Integration of results.

Hypothesis	Quantitative result	Qualitative theme	Interpretation
GenAI → Domain Skills	Significant (+)	AI facilitates skill development	AI supports feedback, practice, and creativity
GenAI → Dependence	Significant (+)	Cognitive offloading	Overuse is associated with less independent thinking
Domain Skills → Performance	Significant (+)	Internalized competence	Skills are associated with confident and accurate performance
Dependence → Performance	Significant (-)	Over-reliance and autonomy	Dependence is associated with weaker performance resilience
Moderation (AI Literacy)	Significant	AI literacy as regulator	AI literacy enables strategic use

offloading related to a lack of engagement in self-regulated learning. The explanation for this is well described by CLT (Sweller, 1988): technology can diminish extraneous cognitive load and also lessen germane load, which leaves less opportunity for schema construction and procedural memory retention. The juxtaposition of Gen-AI being a learning aid and a learning dependency also offers something intriguing — Gen-AI may become a scaffold and a crutch: an enhancement to learning efficiency and a potential threat to long-term learning autonomy. This means thinking about the use of Gen-AI not just as a binary of use or harm, but the context, tool design, and variation in learners' associations with Gen-AI.

The findings also revealed a strong positive link between domain skills related to music and music performance quality, and a strong negative link between reliance on AI tools and music performance quality, indicating two separate dimensions of the relationship between Gen-AI and musical results. Deliberate practice, technical skill and aural ability are key factors in music education and expertise literature that are associated with good performance (Zimmerman & Schunk, 2012). Strong internal skills, including sight-reading, harmonic analysis, and expressive phrasing, are associated with accurate, confident, and artistic performance, even in pressured or unstructured settings. At the same time, part of the strength of the observed association between domain skills and performance quality may reflect same-source measurement, as both constructs captured learners' self-perceptions; this possibility is addressed further in the limitations. The results showed that quality of performance is related to technical skills, expressive interpretation and confidence in performance, based on self-regulatory factors (SRL) and underlying cognitive schemas (CLT). This negative correlation between dependence and performance suggests that overreliance on AI tools could potentially have a negative impact on musical independence and resilience. Students who consistently use AI to write, improvise, and correct their own musical work may lack the development of significant musical schemas and may be more susceptible to performance anxiety, lack of self-efficacy, and inflexibility in the event that AI is not available (Zimmerman, 2002). This two-way effect suggests that Gen-AI indirectly impacts performance via its evolving connection with technology.

The mediation results further indicate that music-specific domain skills and reliance on AI technology have opposing psychological mechanisms that mediate the relationship between Gen-AI use and performance quality, highlighting two conflicting pathways. This positive indirect impact via domain skills indicates that there is an indirect relationship between Gen-AI and improved performance, through the acquisition of musical domain skills including aural skill, technical skill, and theoretical capability. This is consistent with the principles of CLT (Sweller, 2010): AI support designed to reduce extraneous cognitive load can free capacity for more expressive engagement with musical processes, facilitating schema construction and skill internalization, especially when deployed strategically. However, the indirect and negative relationship implies a hidden cost: If the students are using AI for creative and technical work, they might not be as effective at working independently, resulting in lower performance. This pathway is consistent with the impairment of SRL (Zimmerman, 2002) in which the use of AI tools takes the place of the active metacognitive processes of goal setting, self-monitoring, and corrective action. The opposing and indirect connections between these two aspects reflect the ambivalent aspect of Gen-AI in music education: a technology can empower or disempower based on how it is applied. This dualistic perspective is a more comprehensive explanation of the contribution of AI to creative education than the monadic one that focuses on either the benefits or the drawbacks of the technology.

The moderation analysis shows that AI literacy is an important factor shaping the nature and strength of the association of Gen-AI with the music learning process. More specifically, AI literacy enhances the positive relationship between the use of Gen-AI and domain skills related to music and reduces the positive relationship between the use of Gen-AI and dependency on the AI tools. This places AI literacy not just as

a technical skill, but as a metacognitive and regulatory tool that can enhance students' access to the benefits of AI without exposing them to its potential dangers, aligning with its use as a boundary condition rather than an explanatory mechanism. In SRL (Zimmerman, 2002), more AI-literate learners will be more likely to formulate deliberate learning goals, assess AI-generated learning content and feedback, and learn from content that leads to competence-building learning episodes. They are aware of how AI works and its constraints and apply it smartly with AI being a handy tool and not an unquestioned answer. This metacognitive functioning contributes to the germane cognitive load that is needed for deep learning (Sweller, 2010). Meanwhile, AI literacy can help reduce dependency by fostering independence and positive self-efficacy; when students grasp that AI is probabilistic and built on data, they are less inclined to use AI without questioning. The findings highlight the importance of the learner's AI literacy, rather than the technology itself, on the outcome of an AI integration. This makes AI literacy an integral component of music education rather than a mere add-on, and an essential part of the musical skills and practices of the 21st century, empowering students to engage with the swift digitization process with sensitivity, creativity, and competence.

6. Implications

The findings of this study offer important theoretical and practical implications for understanding AI integration in music learning from a psychological perspective. Theoretically, the study extends the application of SRL theory (Zimmerman, 2002) by illustrating that generative AI may serve not only as a learning support tool but also as an external regulatory agent associated with changes in learners' self-regulatory processes. The findings illustrate that generative AI can simultaneously enhance and disrupt the SRL cycle in two distinct yet intertwined ways. On the positive side, generative AI can provide immediate feedback and adaptive support for goal-oriented learning and skill acquisition. However, over-reliance on generative AI represents a breakdown of the SRL cycle, as students lose the self-monitoring, self-evaluating, and strategy-control elements of the process. This indicates that the way students currently use generative AI reflects an emerging psychological reality in which the SRL process is increasingly co-constructed between the student and intelligent technology. From the perspective of CLT (Sweller, 1988), the study adds to understanding of how generative AI interacts with learners' cognitive load during learning and performance. In the positive pathway, Gen-AI reduces extraneous cognitive load by supporting technical execution and procedural fluency, enabling greater allocation of cognitive resources to schema construction and expressive interpretation (germane load). However, in the negative pathway, excessive reliance on AI reduces learners' engagement in essential cognitive processing, limiting schema development and weakening long-term cognitive automation. This dual pattern qualifies the assumption that digital tools decrease cognitive load uniformly across all learners and suggests that generative-AI learning environments may produce conditional cognitive load depending on learners' literacy and usage.

A further contribution is in advancing the emerging literature on technology-mediated cognition. Building on SRL and CLT, the study offers a dual-mechanism account of learning with generative AI in music education, comprising a skill-enhancing pathway (through domain skill development) and a skill-weakening pathway (through dependence). This account moves beyond binary “beneficial versus harmful” views of educational technology by highlighting the psychological tension between cognitive scaffolding and cognitive offloading. The extent to which the learner is capable of using AI technology is crucial in deciding whether the technology becomes a scaffold for learning, or a substitute for cognitive engagement. From this perspective, AI literacy can be viewed as an integral part of SRL-related metacognitive skills, influencing the way learners manage their cognitive resources and exercise independence in AI rich contexts.

From the practical perspective, the integration of AI technology into music education is not only a requirement but also the need to cultivate AI literacy among music learners, particularly in making decisions about the use of AI. Music educators should design learning experiences that promote the appropriate application of AI in ways that keep students in a process of active thought, rather than merely generating content with AI. Structured AI literacy education, focusing on the assessment of AI-generated content, an awareness of AI limitations, and the retention of creative command over musical activities, must be integrated into the curriculum for AI literacy. Similarly, teacher education programs should also impart the skills that enable teachers to facilitate students' creative music making while working with AI rather than letting it replace them.

7. Conclusion and limitations

The present study sought to explore two interlinked pathways by which the use of Gen-AI is associated with performance quality in music learning, through the mediation of music-related domain skills and dependency on AI tools, as well as the moderating role of AI literacy. The findings draw from SRL and CLT, showing a dual pathway effect where the use of Gen-AI was strongly linked with increased domain skills, which was strongly linked with increased performance quality, but also with increased dependence, which was strongly linked with decreased performance quality. Importantly, AI literacy was found to strengthen the positive association between Gen-AI usage and domain skills while weakening the association between Gen-AI usage and dependence. Thus, AI literacy emerges as a critical regulating factor in AI-mediated learning. The mixed-methods design combined quantitative analysis of data from a large pool of music students with qualitative insight into the psychological underpinnings of these associations.

Despite its strengths, the study has several limitations. First, the data are self-reported and cross-sectional, which precludes causal inference and raises the possibility of common method bias and subjective inflation, particularly for the performance-quality measure, which captured perceived rather than independently adjudicated performance. Because all constructs were assessed from the same source at the same time, same-source variance may have inflated the observed associations; this concern is most relevant to the notably strong association between domain skills and performance quality, both of which reflect learners' self-perceptions of improvement and may therefore share method-related variance in addition to substantive overlap. Although procedural remedies and Harman's single-factor test were employed, these cannot fully eliminate such concerns; future research should incorporate longitudinal or experimental designs, objective or instructor-rated performance measures, and multi-source data to strengthen causal and measurement validity. Second, the dependence construct was assessed with only three items; while its reliability and validity were adequate in this sample, future work should develop and validate a fuller, multidimensional dependence scale. Third, the sample was drawn exclusively from six elite conservatories in China, a context characterized by highly selective admission, examination-oriented training traditions, and a national educational-technology policy environment (e.g., the Education Informatization 2.0 Action Plan) that actively promotes digital tools. These cultural and institutional features may shape how students engage with and depend on Gen-AI, so the observed associations may not generalize to Western conservatories, community or informal music-learning settings, non-elite institutions, or lower-resource contexts with different pedagogical traditions and levels of technology access. Cross-cultural and multi-context replication is therefore needed before the model is generalized, and the present findings should be read as context-bound to elite Chinese higher-music education. Finally, participants' familiarity with Gen-AI varied; although the survey defined and introduced the concept and targeted students familiar with digital music tools, differences in individual experience may have influenced how participants interpreted and responded to items. Accordingly, the findings reflect perceived associations that may be shaped by heterogeneous

levels of understanding and usage of Gen-AI. Future research could address this by adopting objective measures of AI use or by recruiting participants with verified Gen-AI experience.

CRedit authorship contribution statement

Yanran Ren: Writing – original draft, Methodology, Data curation, Conceptualization. **Safeer Ullah Khan:** Writing – review & editing, Software, Formal analysis.

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Appendix A

A. Generative AI Usage in Music Learning

- I frequently use generative AI tools during my music learning activities.
- I use generative AI tools to support my music practice.
- I rely on generative AI tools when working on musical tasks.
- I use generative AI to explore musical ideas or compositions.
- I use generative AI tools to assist in understanding music concepts.
- I use generative AI tools for feedback on my music performance.
- I incorporate generative AI tools into my daily music learning routine.
- I use generative AI tools to support music creation or arrangement tasks.
- I engage with generative AI tools when practicing music-related assignments.

B. Music-Specific Domain Skills

- I perceive an improvement in my musical theoretical understanding.
- I perceive an improvement in my aural (listening) skills in music.
- I perceive an improvement in my technical performance skills.
- I perceive an improvement in my ability to interpret musical pieces.
- I perceive an improvement in my improvisational or creative music skills.

C. Dependence on AI Tools

- I tend to rely on generative AI tools when solving music-related problems.
- I often prefer using AI tools instead of solving musical tasks independently.
- I feel I would struggle to complete some music tasks without AI assistance.

D. AI Literacy

- I understand how generative AI systems generate musical or text outputs.
- I am aware of the strengths and limitations of generative AI tools in music learning.
- I can critically evaluate AI-generated musical suggestions.
- I know how to use generative AI tools effectively for learning purposes.
- I can distinguish between useful and less reliable outputs from AI tools.

E. Performance Quality

- I perceive an improvement in the overall quality of my musical performance.
- I feel more confident in my musical performance abilities.
- I am satisfied with my current level of musical performance.
- I perform musical tasks with greater accuracy and expression.
- I perceive that my musical outputs meet higher standards than before.

Data availability

Data will be made available on request.

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